

main  
{

int Node \*head = NULL;

Stack

0x777 → \*head → NULL

// then we use insert-at-end function to create a new node.

insert\_at\_end(&head, 5);

// we sent the address of head from stack  
// "0x777" to the function and sent  
// integer data by 5.

---

How the function receives this address?

And how it safely handles it?



```
Void insert-at-end (int_Node **headRef, int newDate)
```

```
{
```

```
// headRef is now receiving the value 0x777  
// (the original head address in the main).
```

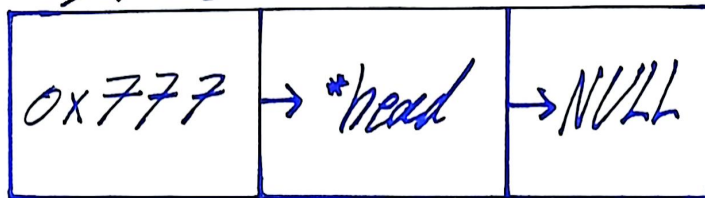
```
if (*headRef == NULL)
```

```
{
```

```
// * headRef means: go to the address  
0x777 and edit the value directly  
inside it. *//
```

Remember:

stack in the main



```
*headRef = node_data (newDate);
```

```
return;
```

```
}
```

```
// The rest of the code
```

```
}
```

Before:

head = NULL

&head = 0x777

headRef = 0x777

\*headRef = NULL (we injection the new address here)

---

After:

head = 0x100 (new address)

&head = 0x777

Scribbling and  
Thin King

~~\*head = NULL  
& head = 0x777~~

0x777

~~\*headRef = NULL  
\*\*headRef = 0x777~~

~~& head~~ ~~\*\*headRef = 0x777~~

headRef = 0x777

~~\*headRef = NULL~~ ~~\*headRef = NULL~~

~~\*\*headRef = 0x777~~

headRef (0x777)

\*headRef = NULL

True